

DIGITAL ELECTRONICS NOTES

Autonomous Engineering Curriculum (JNTUH R18 / R22 Aligned)
Comprehensive 130-Page Stage-1 Project Course Material

UNIT - I: Number System and Boolean Algebra

Number Systems, Base Conversion Methods, Complements of Numbers, Codes - Binary Codes, Binary Coded Decimal Code and its Properties, Unit Distance Codes, Error Detecting and Correcting Codes. Digital Logic Gates (AND, NAND, OR, NOR, EX-OR, EX-NOR), Properties of XOR Gates, Universal Gates, Basic Theorems and Properties, Switching Functions, Canonical and Standard Form.

UNIT - II: Minimization Techniques

Introduction, The minimization with theorems, The Karnaugh Map Method, Three, Four and Five variable K-Maps, Prime and Essential Implications, Don't Care Map Entries, Using the Maps for Simplifying, Quine-McCluskey Method, Multilevel NAND/NOR realizations.

UNIT - III: Combinational Circuits

Design Procedure – Half Adder, Full Adder, Half Subtractor, Full Subtractor, Parallel Binary Adder, Parallel binary subtractor, Binary Multiplier, Multiplexers/DeMultiplexers, decoder, Encoder, Code Converters, Magnitude Comparator. Classification of sequential circuits, The binary cell, The S-R-Latch Flip-Flop, The D-Latch Flip-Flop, The "Clocked T" Flip-Flop, The "Clocked J-K" Flip-Flop, Design of a Clocked Flip-Flop, Timing and Triggering Considerations.

UNIT - IV: Sequential Circuits

Introduction, Basic Architectural Distinctions between Combinational and Sequential circuits, Latches, Flip-Flops, SR, JK, D, T and Master slave, characteristic Tables and equations, Conversion from one type to another, Counters - Design of Single Mode Counter, Ripple Counter, Ring Counter, Shift Register, Ring counter using Shift Register.

UNIT - V: Memory Devices

Classification of memories – ROM : ROM organization, PROM, EPROM, EEPROM, RAM: RAM organization, Write operation, Read operation, Static RAM, Dynamic RAM, Programmable Logic Devices: Programmable Logic Array (PLA), Programmable Array Logic (PAL), Implementation of Combinational Logic using ROM, PLA, PAL.

TEXT BOOKS:

1. Digital Design - Morris Mano, PHI, 3rd Edition.
2. Switching Theory and Logic Design - A. Anand Kumar, PHI, 2nd Edition.
3. Switching and Finite Automata Theory - Zvi Kohavi & Niraj K. Jha, 3rd Edition, Cambridge.

REFERENCE BOOKS:

1. Introduction to Switching Theory and Logic Design – Fredriac J. Hill, Gerald R. Peterson, 3rd Ed, John Wiley & Sons.
2. Digital Fundamentals – A Systems Approach – Thomas L. Floyd, Pearson, 2013.
3. Switching Theory and Logic Design – Bhanu Bhaskara – Tata McGraw Hill Publication, 2012.
4. Fundamentals of Logic Design - Charles H. Roth, Cengage Learning, 5th Edition, 2004.
5. Digital Logic Applications and Design - John M. Yarbrough, Thomson Publications, 2006.
6. Digital Logic and State Machine Design – Comer, 3rd, Oxford, 2013.

COURSE OUTCOMES:

Upon completion of the course, student should possess the following skills:

- Be able to manipulate numeric information in different forms.
- Be able to manipulate simple Boolean expressions using theorems and postulates of Boolean algebra.
- Be able to design and analyze small combinational circuits and standard combinational functions.
- Be able to design and analyze small sequential circuits and standard sequential functions.

COURSE INDEX TABLE

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Total Pages in Document: 130 Pages

Includes Complete Lecture Notes, Derivations, Truth Tables, K-Maps and Logic Realizations.

INTRODUCTION ABOUT DIGITAL SYSTEM

A Digital system is an interconnection of digital modules and it is a system that manipulates discrete elements of information that is represented internally in the binary form.

Now a day's digital systems are used in wide variety of industrial and consumer products such as automated industrial machinery, pocket calculators, microprocessors, digital computers, digital watches, TV games and signal processing and so on.

Characteristics of Digital systems:

- Digital systems manipulate discrete elements of information (digits 0-9 or 26 letters).
- Signals have two discrete values and are therefore said to be binary.
- A signal in digital system represents one binary digit called a bit (0 or 1).

Analog systems vs Digital systems:

- Analog systems process information that varies continuously across time.
- Digital systems use digital circuits that process discrete binary signals (0 or 1).

Advantages of Digital system over Analog system:

1. Ease of programmability: Change program without additional changes in hardware.
2. Reduction in cost of hardware: Enabled by advances in IC technology.
3. High speed: High speed of operation due to Digital Signal Processing.
4. High Reliability: High reliability due to error detection and correction codes.
5. Design is easy: Use of Boolean algebra and digital minimization techniques.
6. Result can be reproduced easily: Output is independent of temperature and noise.

Disadvantages of Digital Systems:

- Use more energy than analog circuits to accomplish same tasks.
- Fragile: single bit error can alter large blocks of related data.
- Quantization error during analog signal sampling.

NUMBER SYSTEM

Number system is a basis for counting items. Modern computers communicate and operate with binary numbers which use only digits 0 & 1. Basic human system is Decimal.

Example: Decimal 18 is represented in binary as 10010.

Three other number systems:

- i) Octal number systems (base 8)
- ii) Hexadecimal number system (base 16)
- iii) Binary Coded Decimal (BCD) system

Binary number system has radix $r = 2$. Weights are powers of 2.

Leftmost bit = Most Significant Bit (MSB); Rightmost bit = Least Significant Bit (LSB).

Solved Example & Equivalence Table:

$$1001.01_2 = [1 \cdot 2^3] + [0 \cdot 2^2] + [0 \cdot 2^1] + [1 \cdot 2^0] + [0 \cdot 2^{-1}] + [1 \cdot 2^{-2}]$$
$$= [1 \cdot 8] + [0 \cdot 4] + [0 \cdot 2] + [1 \cdot 1] + [0 \cdot 0.5] + [1 \cdot 0.25] = 9.25_{10}$$

Decimal | Binary | Octal | Hexadecimal

0		0000		0		0
1		0001		1		1
2		0010		2		2
3		0011		3		3
4		0100		4		4
5		0101		5		5
6		0110		6		6
7		0111		7		7
8		1000		10		8
9		1001		11		9
10		1010		12		A
11		1011		13		B
12		1100		14		C
13		1101		15		D
14		1110		16		E
15		1111		17		F

Number Base Conversions:

i) Binary to Octal: 001 010 011 000 100 101 110 111 -> 1 2 3 0 4 5 6 7₈

ii) Binary to Hexadecimal: 0001 0010 0100 1000 1001 1010 1101 1111 -> 1 2 4 8 9 A D F_{16}

iii) Octal to Binary: 653₈ -> 6(110) 5(101) 3(011) = 110101011₂

iv) Octal to Decimal: $4057.06_8 = 4 \cdot 8^3 + 0 \cdot 8^2 + 5 \cdot 8^1 + 7 \cdot 8^0 + 0 \cdot 8^{-1} + 6 \cdot 8^{-2} = 2095.0937_{10}$

Decimal to Octal & Hexadecimal Conversions:

Convert 378.93_{10} to octal: $378/8 \rightarrow 572_8$; $0.93 \cdot 8 \rightarrow .7341_8 \Rightarrow 572.7341_8$

Convert $5C7_{16}$ to decimal: $5 \cdot 16^2 + 12 \cdot 16^1 + 7 \cdot 16^0 = 1280 + 192 + 7 = 1479_{10}$

Convert 2598.675_{10} to hexadecimal: $2598/16 \rightarrow A26$; $0.675 \cdot 16 \rightarrow .ACCC \Rightarrow A26.ACCC_{16}$

UNIT - 1 : Complements & Signed Binary Arithmetic

Detailed academic lecture notes corresponding to Page no. 7 of the textbook.

Topics covered: Complements & Signed Binary Arithmetic.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Complements & Signed Binary Arithmetic

Detailed academic lecture notes corresponding to Page no. 8 of the textbook.

Topics covered: Complements & Signed Binary Arithmetic.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Complements & Signed Binary Arithmetic

Detailed academic lecture notes corresponding to Page no. 9 of the textbook.

Topics covered: Complements & Signed Binary Arithmetic.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Complements & Signed Binary Arithmetic

Detailed academic lecture notes corresponding to Page no. 10 of the textbook.

Topics covered: Complements & Signed Binary Arithmetic.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Complements & Signed Binary Arithmetic

Detailed academic lecture notes corresponding to Page no. 11 of the textbook.

Topics covered: Complements & Signed Binary Arithmetic.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Complements & Signed Binary Arithmetic

Detailed academic lecture notes corresponding to Page no. 12 of the textbook.

Topics covered: Complements & Signed Binary Arithmetic.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Binary Codes, BCD, Excess-3 & Gray Code

Detailed academic lecture notes corresponding to Page no. 13 of the textbook.

Topics covered: Binary Codes, BCD, Excess-3 & Gray Code.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Binary Codes, BCD, Excess-3 & Gray Code

Detailed academic lecture notes corresponding to Page no. 14 of the textbook.

Topics covered: Binary Codes, BCD, Excess-3 & Gray Code.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Binary Codes, BCD, Excess-3 & Gray Code

Detailed academic lecture notes corresponding to Page no. 15 of the textbook.

Topics covered: Binary Codes, BCD, Excess-3 & Gray Code.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Binary Codes, BCD, Excess-3 & Gray Code

Detailed academic lecture notes corresponding to Page no. 16 of the textbook.

Topics covered: Binary Codes, BCD, Excess-3 & Gray Code.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Binary Codes, BCD, Excess-3 & Gray Code

Detailed academic lecture notes corresponding to Page no. 17 of the textbook.

Topics covered: Binary Codes, BCD, Excess-3 & Gray Code.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Binary Codes, BCD, Excess-3 & Gray Code

Detailed academic lecture notes corresponding to Page no. 18 of the textbook.

Topics covered: Binary Codes, BCD, Excess-3 & Gray Code.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Binary Codes, BCD, Excess-3 & Gray Code

Detailed academic lecture notes corresponding to Page no. 19 of the textbook.

Topics covered: Binary Codes, BCD, Excess-3 & Gray Code.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Binary Codes, BCD, Excess-3 & Gray Code

Detailed academic lecture notes corresponding to Page no. 20 of the textbook.

Topics covered: Binary Codes, BCD, Excess-3 & Gray Code.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Error-Detecting & Correcting Codes (Hamming Codes)

Detailed academic lecture notes corresponding to Page no. 21 of the textbook.

Topics covered: Error-Detecting & Correcting Codes (Hamming Codes).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Error-Detecting & Correcting Codes (Hamming Codes)

Detailed academic lecture notes corresponding to Page no. 22 of the textbook.

Topics covered: Error-Detecting & Correcting Codes (Hamming Codes).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Error-Detecting & Correcting Codes (Hamming Codes)

Detailed academic lecture notes corresponding to Page no. 23 of the textbook.

Topics covered: Error-Detecting & Correcting Codes (Hamming Codes).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Error-Detecting & Correcting Codes (Hamming Codes)

Detailed academic lecture notes corresponding to Page no. 24 of the textbook.

Topics covered: Error-Detecting & Correcting Codes (Hamming Codes).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Error-Detecting & Correcting Codes (Hamming Codes)

Detailed academic lecture notes corresponding to Page no. 25 of the textbook.

Topics covered: Error-Detecting & Correcting Codes (Hamming Codes).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Digital Logic Gates & Universal Logic

Detailed academic lecture notes corresponding to Page no. 26 of the textbook.

Topics covered: Digital Logic Gates & Universal Logic.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Digital Logic Gates & Universal Logic

Detailed academic lecture notes corresponding to Page no. 27 of the textbook.

Topics covered: Digital Logic Gates & Universal Logic.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Boolean Algebra, Huntington Postulates & Canonical Forms

Detailed academic lecture notes corresponding to Page no. 28 of the textbook.

Topics covered: Boolean Algebra, Huntington Postulates & Canonical Forms.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Boolean Algebra, Huntington Postulates & Canonical Forms

Detailed academic lecture notes corresponding to Page no. 29 of the textbook.

Topics covered: Boolean Algebra, Huntington Postulates & Canonical Forms.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Boolean Algebra, Huntington Postulates & Canonical Forms

Detailed academic lecture notes corresponding to Page no. 30 of the textbook.

Topics covered: Boolean Algebra, Huntington Postulates & Canonical Forms.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Boolean Algebra, Huntington Postulates & Canonical Forms

Detailed academic lecture notes corresponding to Page no. 31 of the textbook.

Topics covered: Boolean Algebra, Huntington Postulates & Canonical Forms.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Boolean Algebra, Huntington Postulates & Canonical Forms

Detailed academic lecture notes corresponding to Page no. 32 of the textbook.

Topics covered: Boolean Algebra, Huntington Postulates & Canonical Forms.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 1 : Boolean Algebra, Huntington Postulates & Canonical Forms

Detailed academic lecture notes corresponding to Page no. 33 of the textbook.

Topics covered: Boolean Algebra, Huntington Postulates & Canonical Forms.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
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UNIT - 1 : Boolean Algebra, Huntington Postulates & Canonical Forms

Detailed academic lecture notes corresponding to Page no. 34 of the textbook.

Topics covered: Boolean Algebra, Huntington Postulates & Canonical Forms.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
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- **Practical circuit design principles and solved examination numericals.**

DIGITAL LOGIC DESIGN : Lecture Topic Page 39

Detailed academic lecture notes corresponding to Page no. 35 of the textbook.

Topics covered: Lecture Topic Page 39.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

DIGITAL LOGIC DESIGN : Lecture Topic Page 40

Detailed academic lecture notes corresponding to Page no. 36 of the textbook.

Topics covered: Lecture Topic Page 40.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

DIGITAL LOGIC DESIGN : Lecture Topic Page 41

Detailed academic lecture notes corresponding to Page no. 37 of the textbook.

Topics covered: Lecture Topic Page 41.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
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DIGITAL LOGIC DESIGN : Lecture Topic Page 42

Detailed academic lecture notes corresponding to Page no. 38 of the textbook.

Topics covered: Lecture Topic Page 42.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 2 : Two & Three-Variable Karnaugh Maps (SOP / POS)

Detailed academic lecture notes corresponding to Page no. 39 of the textbook.

Topics covered: Two & Three-Variable Karnaugh Maps (SOP / POS).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 2 : Two & Three-Variable Karnaugh Maps (SOP / POS)

Detailed academic lecture notes corresponding to Page no. 40 of the textbook.

Topics covered: Two & Three-Variable Karnaugh Maps (SOP / POS).

- **Step-by-step mathematical formulation and proofs.**
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UNIT - 2 : Two & Three-Variable Karnaugh Maps (SOP / POS)

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- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 2 : Four, Five & Six-Variable K-Maps, Don't Cares & Prime Implicants

Detailed academic lecture notes corresponding to Page no. 44 of the textbook.

Topics covered: Four, Five & Six-Variable K-Maps, Don't Cares & Prime Implicants.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 2 : Four, Five & Six-Variable K-Maps, Don't Cares & Prime Implicants

Detailed academic lecture notes corresponding to Page no. 45 of the textbook.

Topics covered: Four, Five & Six-Variable K-Maps, Don't Cares & Prime Implicants.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 2 : Four, Five & Six-Variable K-Maps, Don't Cares & Prime Implicants

Detailed academic lecture notes corresponding to Page no. 46 of the textbook.

Topics covered: Four, Five & Six-Variable K-Maps, Don't Cares & Prime Implicants.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 2 : Four, Five & Six-Variable K-Maps, Don't Cares & Prime Implicants

Detailed academic lecture notes corresponding to Page no. 47 of the textbook.

Topics covered: Four, Five & Six-Variable K-Maps, Don't Cares & Prime Implicants.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 2 : Four, Five & Six-Variable K-Maps, Don't Cares & Prime Implicants

Detailed academic lecture notes corresponding to Page no. 48 of the textbook.

Topics covered: Four, Five & Six-Variable K-Maps, Don't Cares & Prime Implicants.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 2 : Four, Five & Six-Variable K-Maps, Don't Cares & Prime Implicants

Detailed academic lecture notes corresponding to Page no. 49 of the textbook.

Topics covered: Four, Five & Six-Variable K-Maps, Don't Cares & Prime Implicants.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 2 : Four, Five & Six-Variable K-Maps, Don't Cares & Prime Implicants

Detailed academic lecture notes corresponding to Page no. 50 of the textbook.

Topics covered: Four, Five & Six-Variable K-Maps, Don't Cares & Prime Implicants.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 2 : Quine-McCluskey (Tabular) Minimization Method

Detailed academic lecture notes corresponding to Page no. 51 of the textbook.

Topics covered: Quine-McCluskey (Tabular) Minimization Method.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 2 : Quine-McCluskey (Tabular) Minimization Method

Detailed academic lecture notes corresponding to Page no. 52 of the textbook.

Topics covered: Quine-McCluskey (Tabular) Minimization Method.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 2 : Quine-McCluskey (Tabular) Minimization Method

Detailed academic lecture notes corresponding to Page no. 53 of the textbook.

Topics covered: Quine-McCluskey (Tabular) Minimization Method.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 2 : Quine-McCluskey (Tabular) Minimization Method

Detailed academic lecture notes corresponding to Page no. 54 of the textbook.

Topics covered: Quine-McCluskey (Tabular) Minimization Method.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 2 : Quine-McCluskey (Tabular) Minimization Method

Detailed academic lecture notes corresponding to Page no. 55 of the textbook.

Topics covered: Quine-McCluskey (Tabular) Minimization Method.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 2 : Quine-McCluskey (Tabular) Minimization Method

Detailed academic lecture notes corresponding to Page no. 56 of the textbook.

Topics covered: Quine-McCluskey (Tabular) Minimization Method.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 2 : Quine-McCluskey (Tabular) Minimization Method

Detailed academic lecture notes corresponding to Page no. 57 of the textbook.

Topics covered: Quine-McCluskey (Tabular) Minimization Method.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 2 : Quine-McCluskey (Tabular) Minimization Method

Detailed academic lecture notes corresponding to Page no. 58 of the textbook.

Topics covered: Quine-McCluskey (Tabular) Minimization Method.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 2 : Quine-McCluskey (Tabular) Minimization Method

Detailed academic lecture notes corresponding to Page no. 59 of the textbook.

Topics covered: Quine-McCluskey (Tabular) Minimization Method.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Combinational Logic: Half/Full Adders & Subtractors

Detailed academic lecture notes corresponding to Page no. 60 of the textbook.

Topics covered: Combinational Logic: Half/Full Adders & Subtractors.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Combinational Logic: Half/Full Adders & Subtractors

Detailed academic lecture notes corresponding to Page no. 61 of the textbook.

Topics covered: Combinational Logic: Half/Full Adders & Subtractors.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Combinational Logic: Half/Full Adders & Subtractors

Detailed academic lecture notes corresponding to Page no. 62 of the textbook.

Topics covered: Combinational Logic: Half/Full Adders & Subtractors.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Combinational Logic: Half/Full Adders & Subtractors

Detailed academic lecture notes corresponding to Page no. 63 of the textbook.

Topics covered: Combinational Logic: Half/Full Adders & Subtractors.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Combinational Logic: Half/Full Adders & Subtractors

Detailed academic lecture notes corresponding to Page no. 64 of the textbook.

Topics covered: Combinational Logic: Half/Full Adders & Subtractors.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Combinational Logic: Half/Full Adders & Subtractors

Detailed academic lecture notes corresponding to Page no. 65 of the textbook.

Topics covered: Combinational Logic: Half/Full Adders & Subtractors.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Combinational Logic: Half/Full Adders & Subtractors

Detailed academic lecture notes corresponding to Page no. 66 of the textbook.

Topics covered: Combinational Logic: Half/Full Adders & Subtractors.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Combinational Logic: Half/Full Adders & Subtractors

Detailed academic lecture notes corresponding to Page no. 67 of the textbook.

Topics covered: Combinational Logic: Half/Full Adders & Subtractors.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Combinational Logic: Half/Full Adders & Subtractors

Detailed academic lecture notes corresponding to Page no. 68 of the textbook.

Topics covered: Combinational Logic: Half/Full Adders & Subtractors.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Parallel Binary Adders, Ripple Carry & Carry Look-Ahead (CLA)

Detailed academic lecture notes corresponding to Page no. 69 of the textbook.

Topics covered: Parallel Binary Adders, Ripple Carry & Carry Look-Ahead (CLA).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Parallel Binary Adders, Ripple Carry & Carry Look-Ahead (CLA)

Detailed academic lecture notes corresponding to Page no. 70 of the textbook.

Topics covered: Parallel Binary Adders, Ripple Carry & Carry Look-Ahead (CLA).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Parallel Binary Adders, Ripple Carry & Carry Look-Ahead (CLA)

Detailed academic lecture notes corresponding to Page no. 71 of the textbook.

Topics covered: Parallel Binary Adders, Ripple Carry & Carry Look-Ahead (CLA).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Parallel Binary Adders, Ripple Carry & Carry Look-Ahead (CLA)

Detailed academic lecture notes corresponding to Page no. 72 of the textbook.

Topics covered: Parallel Binary Adders, Ripple Carry & Carry Look-Ahead (CLA).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Parallel Binary Adders, Ripple Carry & Carry Look-Ahead (CLA)

Detailed academic lecture notes corresponding to Page no. 73 of the textbook.

Topics covered: Parallel Binary Adders, Ripple Carry & Carry Look-Ahead (CLA).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Parallel Binary Adders, Ripple Carry & Carry Look-Ahead (CLA)

Detailed academic lecture notes corresponding to Page no. 74 of the textbook.

Topics covered: Parallel Binary Adders, Ripple Carry & Carry Look-Ahead (CLA).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : BCD Adder, Excess-3 Adder & Serial Multiplier

Detailed academic lecture notes corresponding to Page no. 75 of the textbook.

Topics covered: BCD Adder, Excess-3 Adder & Serial Multiplier.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : BCD Adder, Excess-3 Adder & Serial Multiplier

Detailed academic lecture notes corresponding to Page no. 76 of the textbook.

Topics covered: BCD Adder, Excess-3 Adder & Serial Multiplier.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : BCD Adder, Excess-3 Adder & Serial Multiplier

Detailed academic lecture notes corresponding to Page no. 77 of the textbook.

Topics covered: BCD Adder, Excess-3 Adder & Serial Multiplier.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : BCD Adder, Excess-3 Adder & Serial Multiplier

Detailed academic lecture notes corresponding to Page no. 78 of the textbook.

Topics covered: BCD Adder, Excess-3 Adder & Serial Multiplier.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : BCD Adder, Excess-3 Adder & Serial Multiplier

Detailed academic lecture notes corresponding to Page no. 79 of the textbook.

Topics covered: BCD Adder, Excess-3 Adder & Serial Multiplier.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Code Converters, Comparators, Encoders & Tristate Bus

Detailed academic lecture notes corresponding to Page no. 80 of the textbook.

Topics covered: Code Converters, Comparators, Encoders & Tristate Bus.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Code Converters, Comparators, Encoders & Tristate Bus

Detailed academic lecture notes corresponding to Page no. 81 of the textbook.

Topics covered: Code Converters, Comparators, Encoders & Tristate Bus.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Code Converters, Comparators, Encoders & Tristate Bus

Detailed academic lecture notes corresponding to Page no. 82 of the textbook.

Topics covered: Code Converters, Comparators, Encoders & Tristate Bus.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Code Converters, Comparators, Encoders & Tristate Bus

Detailed academic lecture notes corresponding to Page no. 83 of the textbook.

Topics covered: Code Converters, Comparators, Encoders & Tristate Bus.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Code Converters, Comparators, Encoders & Tristate Bus

Detailed academic lecture notes corresponding to Page no. 84 of the textbook.

Topics covered: Code Converters, Comparators, Encoders & Tristate Bus.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Code Converters, Comparators, Encoders & Tristate Bus

Detailed academic lecture notes corresponding to Page no. 85 of the textbook.

Topics covered: Code Converters, Comparators, Encoders & Tristate Bus.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Code Converters, Comparators, Encoders & Tristate Bus

Detailed academic lecture notes corresponding to Page no. 86 of the textbook.

Topics covered: Code Converters, Comparators, Encoders & Tristate Bus.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Code Converters, Comparators, Encoders & Tristate Bus

Detailed academic lecture notes corresponding to Page no. 87 of the textbook.

Topics covered: Code Converters, Comparators, Encoders & Tristate Bus.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 3 : Code Converters, Comparators, Encoders & Tristate Bus

Detailed academic lecture notes corresponding to Page no. 88 of the textbook.

Topics covered: Code Converters, Comparators, Encoders & Tristate Bus.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Latches, Flip-Flops (SR, JK, D, T, Master-Slave) & Excitation Tables

Detailed academic lecture notes corresponding to Page no. 89 of the textbook.

Topics covered: Latches, Flip-Flops (SR, JK, D, T, Master-Slave) & Excitation Tables.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Latches, Flip-Flops (SR, JK, D, T, Master-Slave) & Excitation Tables

Detailed academic lecture notes corresponding to Page no. 90 of the textbook.

Topics covered: Latches, Flip-Flops (SR, JK, D, T, Master-Slave) & Excitation Tables.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Latches, Flip-Flops (SR, JK, D, T, Master-Slave) & Excitation Tables

Detailed academic lecture notes corresponding to Page no. 91 of the textbook.

Topics covered: Latches, Flip-Flops (SR, JK, D, T, Master-Slave) & Excitation Tables.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Latches, Flip-Flops (SR, JK, D, T, Master-Slave) & Excitation Tables

Detailed academic lecture notes corresponding to Page no. 92 of the textbook.

Topics covered: Latches, Flip-Flops (SR, JK, D, T, Master-Slave) & Excitation Tables.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Latches, Flip-Flops (SR, JK, D, T, Master-Slave) & Excitation Tables

Detailed academic lecture notes corresponding to Page no. 93 of the textbook.

Topics covered: Latches, Flip-Flops (SR, JK, D, T, Master-Slave) & Excitation Tables.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Latches, Flip-Flops (SR, JK, D, T, Master-Slave) & Excitation Tables

Detailed academic lecture notes corresponding to Page no. 94 of the textbook.

Topics covered: Latches, Flip-Flops (SR, JK, D, T, Master-Slave) & Excitation Tables.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Sequential Circuit Design & Shift Registers (Universal 74194)

Detailed academic lecture notes corresponding to Page no. 95 of the textbook.

Topics covered: Sequential Circuit Design & Shift Registers (Universal 74194).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Sequential Circuit Design & Shift Registers (Universal 74194)

Detailed academic lecture notes corresponding to Page no. 96 of the textbook.

Topics covered: Sequential Circuit Design & Shift Registers (Universal 74194).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Sequential Circuit Design & Shift Registers (Universal 74194)

Detailed academic lecture notes corresponding to Page no. 97 of the textbook.

Topics covered: Sequential Circuit Design & Shift Registers (Universal 74194).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Sequential Circuit Design & Shift Registers (Universal 74194)

Detailed academic lecture notes corresponding to Page no. 98 of the textbook.

Topics covered: Sequential Circuit Design & Shift Registers (Universal 74194).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Sequential Circuit Design & Shift Registers (Universal 74194)

Detailed academic lecture notes corresponding to Page no. 99 of the textbook.

Topics covered: Sequential Circuit Design & Shift Registers (Universal 74194).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Sequential Circuit Design & Shift Registers (Universal 74194)

Detailed academic lecture notes corresponding to Page no. 100 of the textbook.

Topics covered: Sequential Circuit Design & Shift Registers (Universal 74194).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Sequential Circuit Design & Shift Registers (Universal 74194)

Detailed academic lecture notes corresponding to Page no. 101 of the textbook.

Topics covered: Sequential Circuit Design & Shift Registers (Universal 74194).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Sequential Circuit Design & Shift Registers (Universal 74194)

Detailed academic lecture notes corresponding to Page no. 102 of the textbook.

Topics covered: Sequential Circuit Design & Shift Registers (Universal 74194).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Sequential Circuit Design & Shift Registers (Universal 74194)

Detailed academic lecture notes corresponding to Page no. 103 of the textbook.

Topics covered: Sequential Circuit Design & Shift Registers (Universal 74194).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Sequential Circuit Design & Shift Registers (Universal 74194)

Detailed academic lecture notes corresponding to Page no. 104 of the textbook.

Topics covered: Sequential Circuit Design & Shift Registers (Universal 74194).

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters

Detailed academic lecture notes corresponding to Page no. 105 of the textbook.

Topics covered: Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters

Detailed academic lecture notes corresponding to Page no. 106 of the textbook.

Topics covered: Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters

Detailed academic lecture notes corresponding to Page no. 107 of the textbook.

Topics covered: Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters

Detailed academic lecture notes corresponding to Page no. 108 of the textbook.

Topics covered: Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters

Detailed academic lecture notes corresponding to Page no. 109 of the textbook.

Topics covered: Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters

Detailed academic lecture notes corresponding to Page no. 110 of the textbook.

Topics covered: Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters

Detailed academic lecture notes corresponding to Page no. 111 of the textbook.

Topics covered: Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters

Detailed academic lecture notes corresponding to Page no. 112 of the textbook.

Topics covered: Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters

Detailed academic lecture notes corresponding to Page no. 113 of the textbook.

Topics covered: Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters

Detailed academic lecture notes corresponding to Page no. 114 of the textbook.

Topics covered: Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters

Detailed academic lecture notes corresponding to Page no. 115 of the textbook.

Topics covered: Counters: Asynchronous (Mod-6, Mod-10) & Synchronous Counters.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Ring Counters & Johnson (Twisted Ring) Counters

Detailed academic lecture notes corresponding to Page no. 116 of the textbook.

Topics covered: Ring Counters & Johnson (Twisted Ring) Counters.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Ring Counters & Johnson (Twisted Ring) Counters

Detailed academic lecture notes corresponding to Page no. 117 of the textbook.

Topics covered: Ring Counters & Johnson (Twisted Ring) Counters.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 4 : Ring Counters & Johnson (Twisted Ring) Counters

Detailed academic lecture notes corresponding to Page no. 118 of the textbook.

Topics covered: Ring Counters & Johnson (Twisted Ring) Counters.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 5 : Classification of Memories: ROM, PROM, EPROM, SRAM & DRAM

Detailed academic lecture notes corresponding to Page no. 119 of the textbook.

Topics covered: Classification of Memories: ROM, PROM, EPROM, SRAM & DRAM.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 5 : Classification of Memories: ROM, PROM, EPROM, SRAM & DRAM

Detailed academic lecture notes corresponding to Page no. 120 of the textbook.

Topics covered: Classification of Memories: ROM, PROM, EPROM, SRAM & DRAM.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 5 : Classification of Memories: ROM, PROM, EPROM, SRAM & DRAM

Detailed academic lecture notes corresponding to Page no. 121 of the textbook.

Topics covered: Classification of Memories: ROM, PROM, EPROM, SRAM & DRAM.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 5 : Memory Decoding & Binary Cell Architecture

Detailed academic lecture notes corresponding to Page no. 122 of the textbook.

Topics covered: Memory Decoding & Binary Cell Architecture.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 5 : Programmable Logic Devices: PLA & PAL Implementations

Detailed academic lecture notes corresponding to Page no. 123 of the textbook.

Topics covered: Programmable Logic Devices: PLA & PAL Implementations.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 5 : Programmable Logic Devices: PLA & PAL Implementations

Detailed academic lecture notes corresponding to Page no. 124 of the textbook.

Topics covered: Programmable Logic Devices: PLA & PAL Implementations.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 5 : Programmable Logic Devices: PLA & PAL Implementations

Detailed academic lecture notes corresponding to Page no. 125 of the textbook.

Topics covered: Programmable Logic Devices: PLA & PAL Implementations.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**

UNIT - 5 : Programmable Logic Devices: PLA & PAL Implementations

Detailed academic lecture notes corresponding to Page no. 126 of the textbook.

Topics covered: Programmable Logic Devices: PLA & PAL Implementations.

- **Step-by-step mathematical formulation and proofs.**
- **Standard Boolean and algebraic derivations matching JNTUH curriculum.**
- **Truth tables, transition equations, and logic gate implementations.**
- **Practical circuit design principles and solved examination numericals.**